# Horticultural Greenhouse Project

Sensors, control and remote monitoring

Goal: monitor temperature, soil moisture and light, then control the greenhouse actuators.

ENSEA - Electronics 1A

Date: 21/05

# What is a greenhouse?

## Definition

A greenhouse protects plants and helps stabilize their growing conditions.

## Key variables

Temperature, soil moisture and light.

## Our project

Add sensors, Bluetooth control and basic actuators to a small prototype.

# Presentation structure



**Part 1**

Design

Need, architecture and prototype.

**Part 2**

Tests

Bluetooth, sensors and actuators.

**Part 3**

PCB and feedback

PCB choices, results and limits.

One project, three technical parts.

# Specifications and PCB strategy



## Main goal

Connected greenhouse: measure, transmit and control.

| Function | Expected solution |
|---|---|
| **Temperature** | DS18B20 + fan / heating command |
| **Moisture** | Capacitive moisture sensor + pump command |
| **Light** | LDR + LED / lighting mode |
| **Remote control** | HM10 Bluetooth + STM32 |
| **Prototype** | 50 x 30 cm greenhouse mockup |

## PCB strategy evolution

1. First idea: one Nucleo shield.
2. Then: split the system into 4 PCBs.
3. Final choice: simplify into 3 PCBs.

## Why this change?

- Easier routing and soldering.
- Lower short-circuit risk.
- Easier block-by-block testing.
- Cleaner separation between sensors, power and commands.

Each function had to be testable before full integration.

# Part 1

Need and system design

What problem are we trying to solve, and how is the system organized?

# Context and specifications



### Project need

Monitor a small greenhouse and trigger simple actions remotely. **Main variables**

**Temperature** **Moisture** **Light**

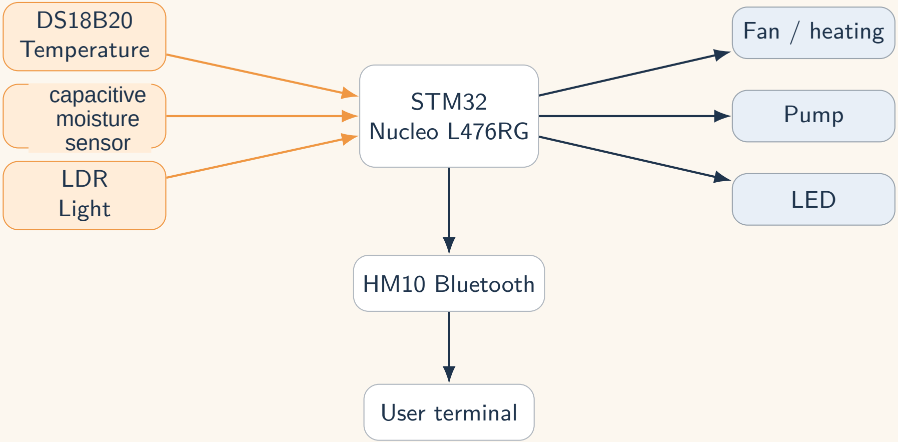
### Expected functions

- Measure values.
- Send them by Bluetooth.
- Control actuators.
- Test each block.

Need $\rightarrow$ choices $\rightarrow$ prototype $\rightarrow$ tests $\rightarrow$ analysis.

# System architecture






The STM32 is the central block: it reads sensors, controls outputs and communicates with the user.

# Sensors selected



### DS18B20

Digital temperature sensor.

Needs a pull-up resistor on the data line.

### Capacitive moisture sensor

Soil moisture sensor.

Calibrated between air and water values.

### LDR

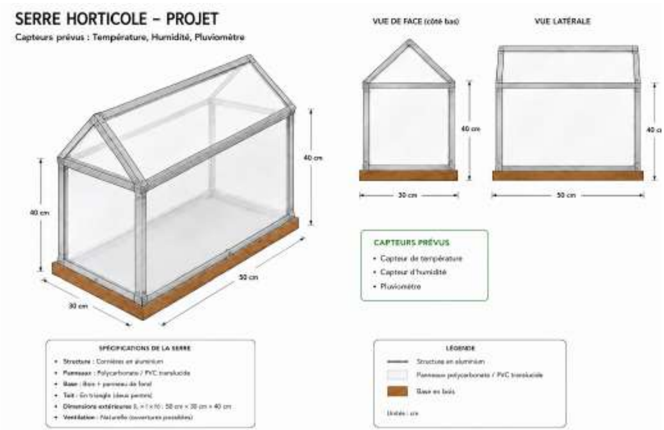
Analog light sensor.

Thresholds chosen after low and high light tests.

Sensor calibration is essential because raw values depend on the test conditions.

# Prototype model






**Assembly time**
**16 h**

### Designed and built by our team

We assembled the greenhouse ourselves, from the base to the structure.

- Wooden base
- PVC / translucent panels
- Pitched roof
- Size: 50 x 30 x 40 cm

# Final prototype built by our team

**Assembly time**
**16 h**

## Manual assembly

The three team members built the prototype ourselves.

- Panel cutting and placement
- Frame and base fixing
- Alignment checks
- Ready for electronics

# Part 2

Control, Bluetooth and tests

How do we command the system and prove that each block works?

# Bluetooth module HM10



Bluetooth link between the smartphone and the STM32.

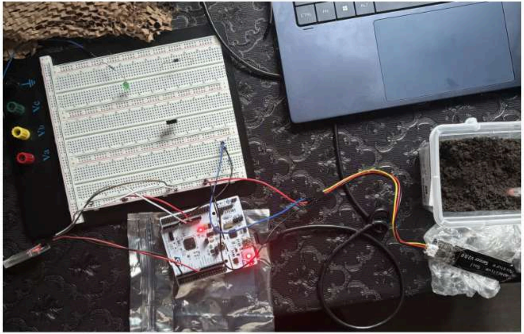
## Role in the project

- Receives user commands.
- Sends sensor values back.
- Makes the system controllable remotely.

## Example commands

TEMP? HUM? FAN ON/OFF

# Capacitive moisture sensor

### Principle of the sensor

Analog value linked to soil moisture.

### Calibration

- Air = dry reference.
- Water = wet reference.
- Thresholds from tests.

The raw value changes with the sensor and the test setup.

# Moisture sensor calibration



**Dry test**

**Wet test**

These two cases define the usable range for the humidity measurement.

# Moisture Bluetooth result



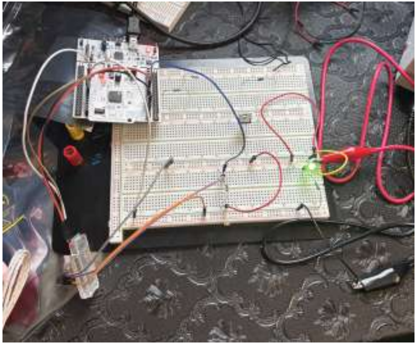

## Validated

- ADC reading
- Bluetooth send
- Remote display

Value received on the terminal.

# Temperature sensor DS18B20





## DS18B20

Digital temperature measurement.

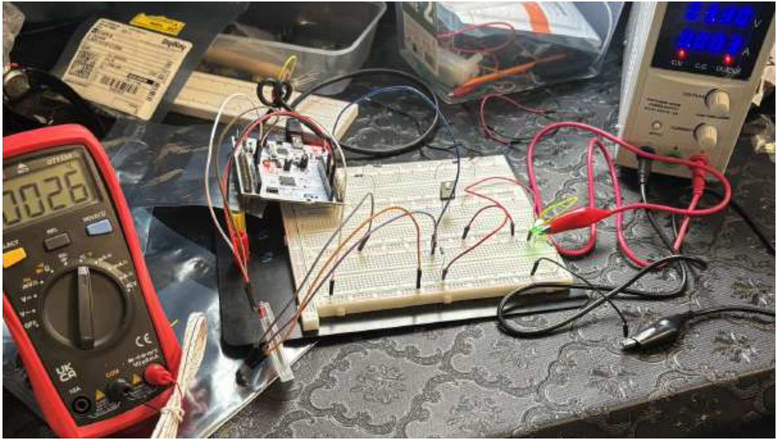
## Wiring

- One data line.
- Pull-up resistor required.
- Value sent by Bluetooth.

Without the pull-up, the data line may become unstable.

# Temperature measurement test





# Temperature Bluetooth response



# Light sensor LDR

Analog light measurement.

### Test method

- Low-light test.
- Strong-light test.
- Compare ADC values.
- Define thresholds.

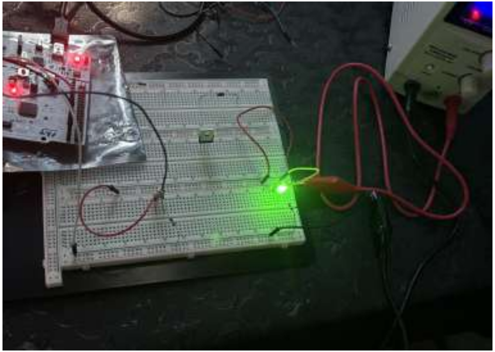
### Use

Control lighting or detect excessive brightness.

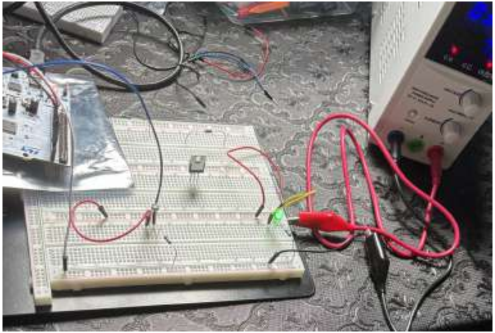
# LDR calibration tests



### Low light test



### Strong light test



# LDR Bluetooth results

## Low light


## Strong light

# LDR thresholds retained



| ADC value | Light level |
|---|---|
| $> 3000$ | Strong light |
| $> 1500$ | Medium light |
| **Otherwise** | Low light |

### Interpretation

These values are experimental thresholds.

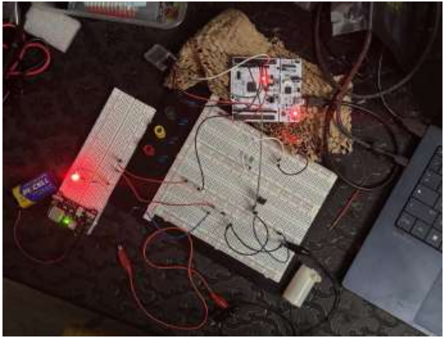
### Next use

The STM32 can apply these thresholds to control a LED or lighting mode.

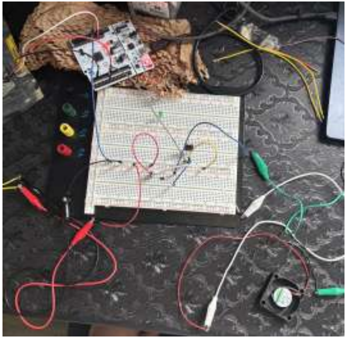
# Pump and fan tests



**Pump test**



**Fan test**



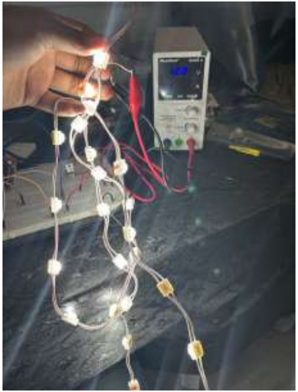
Pump and fan were individually controllable by remote commands.

# Fan Bluetooth result



# Component test summary





**Validated separately**

- Bluetooth.
- Moisture.
- Temperature.
- Light detection.
- Pump and fan.

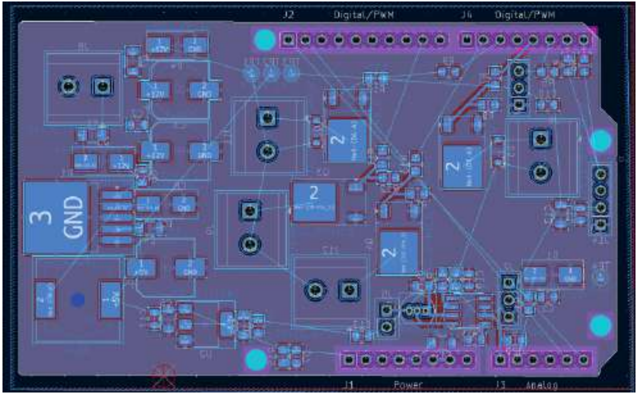
The main difficulty remained full integration.

# Part 3

Prototype, PCB and feedback

What was built, what worked, and what should be improved?

# PCB assembly and connection



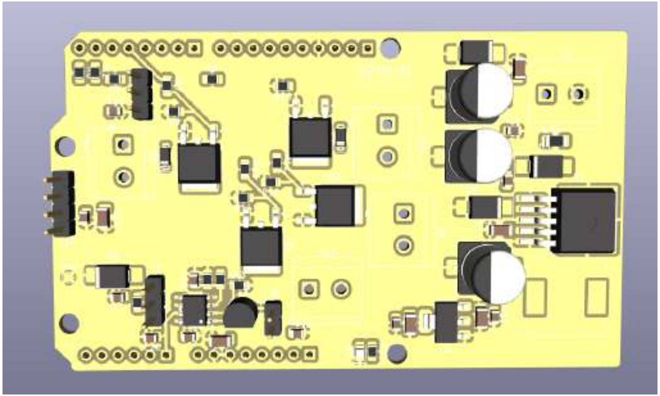

## Integration work

- Solder the components.
- Check power rails.
- Connect sensors and outputs.
- Test each board before full assembly.

The global integration remained the hardest part of the project.

# PCB design: first approach





**Initial idea**

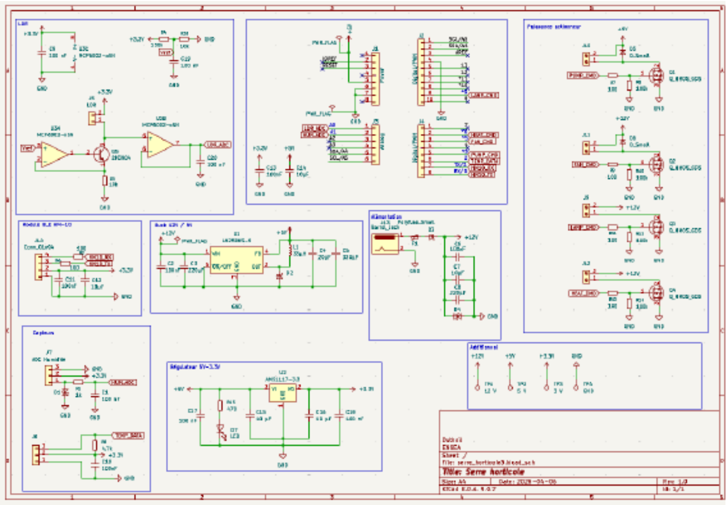
A single shield connected to the STM32 Nucleo.

**Limits**

- Dense routing.
- Harder soldering.
- Higher short-circuit risk.
- Debugging more difficult.

We finally moved away from the single-shield idea to make the system easier to test and repair.

# Complete schematic overview

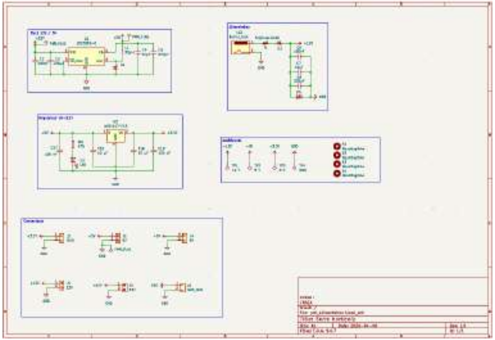


# Electronic schematics: power and sensors



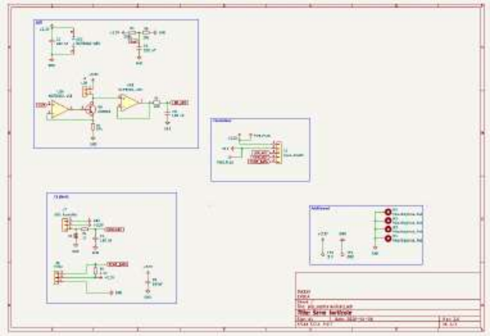
## Power supply board

12 V input, 5 V and 3.3 V rails.



## Sensor board

LDR, ME110 and DS18B20 inputs.

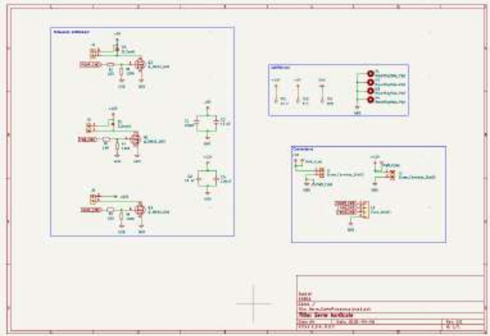


# Electronic schematics: outputs



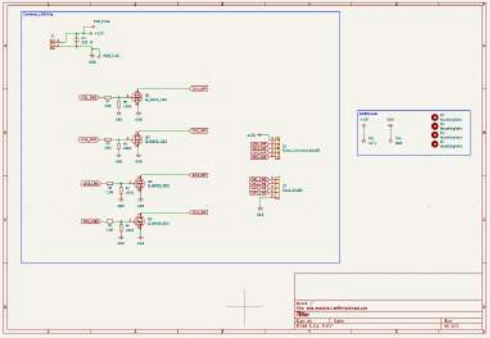
## Actuator board

MOSFET stages for pump, fan and heating.



## LED strip board

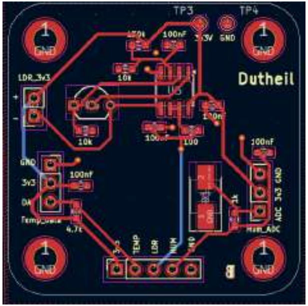
Four MOSFET channels.



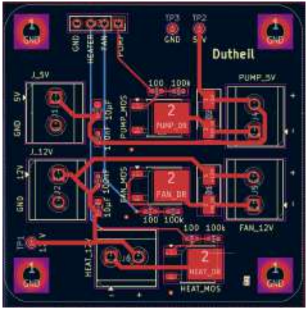
# PCB layout: split into functional boards



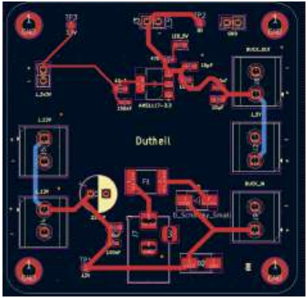
**Sensor board**




**Power board**




**Actuator board**



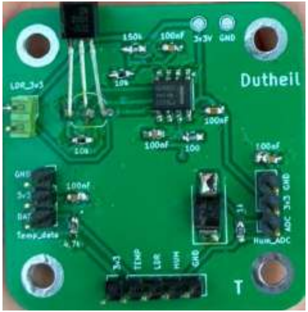

Final strategy: separate sensors, power supply and actuator command stages.

# Fabricated PCB boards

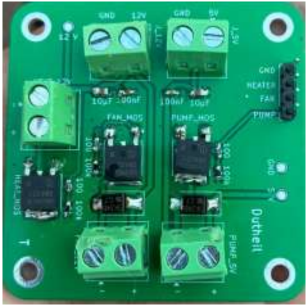


### Power supply

### Sensors




### Actuators




The smaller boards made soldering and block-by-block testing more manageable.

# Results, limits and possible causes



### Validated separately

- Several sensor readings.
- Bluetooth communication.
- Pump and fan commands.
- Mechanical model.

### Main limit

The complete system was not fully reliable.

The exact cause was not identified before the end of the project.

Possible causes: weak solder joint, wiring error, power issue, software timing issue or Bluetooth conflict.

# Conclusion and next steps



**Conclusion**

Several blocks worked separately, but the complete greenhouse was not finalized.

The main difficulty was full system integration.

**Next steps**

- Finish remote controlling.
- Rework the power-supply PCB.
- Stabilize the complete integration.

# Questions?

Thank you for your attention.